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**MEASUREMENT OF THE UNSTEADY THERMAL GREEN'S FUNCTION IN A
BOUNDARY LAYER FLOW: A PRELIMINARY THEORY**

Gregg Radtke

Department of Aerospace and
Mechanical Engineering
The University of Arizona
Tucson, AZ 85721
garadtke@email.arizona.edu

Alfonso Ortega

Department of Aerospace and
Mechanical Engineering
The University of Arizona
Tucson, AZ 85721
ortega@u.arizona.edu

Barry Ganapol

Department of Aerospace and
Mechanical Engineering
The University of Arizona
Tucson, AZ 85721
ganapol@cowboy.ame.arizona.edu

ABSTRACT

Compared to the heat transfer coefficient, the thermal Green's function concept is a more fundamental method of describing the relationship between local wall heat transfer and wall temperature. It is far more amenable to situations involving strong spatial, temporal, and boundary condition variability. The utility of this methodology has been established, in particular for the analysis of the conjugate heat transfer problem. A necessary element in this technique is an inverse theoretical model to infer the Green's function from laboratory thermal response data. This paper presents preliminary results from a first attempt to develop such a measurement theory to extract local approximations to the unsteady thermal Green's function (UTGF) in 3-D boundary layer flows. The flow model used is a linear shear flow, which is a solution valid in the near wall region of laminar flows, as well as the viscous sublayer region of a turbulent boundary layer. This model is governed by the shear velocity, which is a measure of the local wall shear stress. The solution methodology employs the mathematical theory of Green's function solutions to the energy equation for a general 3-D boundary layer flow, where the UTGF of interest is the thermal response to an impulsive heat load. Analytic methods are used to condense the equation from a 3-D to a 2-D transient PDE, and the reduced equation is solved using a Petrov-Galerkin Finite Element Method. These data are used to construct a numerical UTGF uniquely determined by the shear velocity, flow angle, and the thermodynamic properties of the fluid. An error minimization scheme is proposed to find the appropriate value of the shear velocity, thus providing local UTGF, and shear velocity measurements.

Keywords: Green's function, forced convection, boundary layer

NOMENCLATURE

c	fluid specific heat
D	length scale of the heat source
E	error
$\hat{g}$	nondimensional 2-D UTGF
G	global 3-D UTGF
$\hat{G}$	nondimensional 3-D UTGF
k	fluid thermal conductivity
Pe_τ	Peclet number based on the shear velocity
Pr	fluid Prandtl number ($=\nu/\alpha$)
q	boundary heat flux
Q	heat flux scale
T_∞	freestream temperature
T	temperature field
t	time
$\mathbf{u}$	velocity field
u_τ	shear velocity
x	streamwise coordinate
y	spanwise coordinate
z	normal coordinate

Greek Symbols

α	molecular thermal diffusivity
β	shear flow angle
δ	Dirac's delta function
ε	effective thermal diffusivity
ε_H	eddy diffusivity
ζ	local nondimensional normal coordinate
η	local nondimensional spanwise coordinate
θ	nondimensional temperature
μ	fluid viscosity
ν	fluid kinematic viscosity

- ξ local nondimensional streamwise coordinate
- ρ fluid density
- τ nondimensional time
- τ_0 wall shear stress
- σ nondimensional heat flux

Subscript

- M experimentally measured values

INTRODUCTION

The limitations of the heat transfer coefficient as a descriptor for convective heat transfer have been known for some time. In particular, the heat transfer coefficient shows strong spatial and boundary condition dependence in flows of engineering interest, which has led to the search for invariant descriptors for convective heat transfer [1] in order to enable thermal analysis to proceed without relying on full computational fluid dynamics (CFD) solutions. The thermal Green's function is an alternative formulation to the convective heat transfer problem.

The thermal Green's function approach utilizes the superposition of kernel, or point source solutions to the energy equation. This is allowed as long as the thermal properties of the fluid can be considered temperature independent. In cases where the Green's function can be determined, either analytically or numerically, computation of the thermal response for any heat flux boundary condition reduces to ordinary Riemann integration. Thus, the Green's function fulfills a similar role as the heat transfer coefficient by specifying a relationship between heat flux and thermal response, although now as the kernel function in an integral operator rather than a simple constant of proportionality. If the physics of the flow are modeled correctly, the Green's function will be invariant with respect to the thermal boundary condition.

A difficulty inherent in this approach is the availability of Green's functions for the flow field of interest. Analytic kernel solutions are available only for very simplified flow models, and numerical Green's functions require detailed knowledge of the flow field and often considerable computational expense. Moreover, the storage of Green's function data is a considerable issue, particularly for 3-D and/or transient cases. What is needed is a simplified way to characterize the Green's function in terms of a convenient parameterization, and a streamlined methodology to measure such Green's functions in more general flows.

PREVIOUS STUDIES

There are several examples of analytic Green's functions used in convective heat transfer, and the most relevant are discussed here, beginning with those using a steady analytic form of the Green's function for 3-D conjugate heat transfer. Ortega and Ramanathan [2], [3] used the kernel for a uniform

flow effective diffusivity model to predict the thermal response from square sources of heat in electronics cooling applications. The governing parameters for this simplified model are the velocity and "effective" diffusivity, which was interpreted as the sum of molecular and eddy diffusivities. In another study of conjugate heat transfer for electronics cooling applications, Li [4] demonstrated the use of an analytic kernel for a shear flow model neglecting streamwise diffusion in the fluid, where the flow is characterized by a shear velocity, a measure of shear stress at the wall.

Cole and Beck [5] utilized the unsteady surface element method in the analysis of 2-D transient conjugate heat transfer from a strip heater, which used the shear flow model with substantial approximations to the thermal model. Analytic "fundamental" solutions were used, which can be related to Green's functions via integration over a unit step function. Liang and Cole [6] later extended this work for 3-D flow fields.

The discrete Green's function approach [7] represents the first attempt towards direct measurement of the Green's function in a general flow, in this case a discretized version rather than a continuous Green's function. For steady 2-D geometries, this has been demonstrated for cases such as the backward facing step by Hacker and Eaton [8] and turbomachinery applications by Mukerji and Eaton [9]. However, this measurement technique, in principle requires independent measurements of the thermal response to heat sources at more than one location, as well as storage of a matrix of values to represent the Green's function rather than a simple number of parameters. Furthermore, limitations observed by Batchelder and Moffat [10] currently prevent accurately resolving true 3-D flow fields, and there seems to be no procedure to extend this methodology to transient heat transfer.

The objectives of the proposed experimental technique are straightforward. In order to characterize the local wall convective heat transfer from a surface, it is proposed to apply localized heating over a "spot" on the surface, and then optically inspect the transient thermal response. The relevant questions then are: "How does one use the spatial-temporal thermal response to infer the unsteady thermal Green's function (UTGF)? Secondly, how does one use the UTGF to infer the relationship between local wall heat flux and local wall temperature?"

This paper presents a preliminary theory for using transient thermal response data to obtain UTGF measurements. The linear shear flow model is used, which is a near-wall approximation in both a laminar boundary layer or within the viscous sublayer of the turbulent boundary layer. Previous research using this model has focused primarily on steady solutions neglecting streamwise and often spanwise diffusion effects where analytic solutions are available. The present work uses a numerical UTGF determined using a finite element method (FEM).

THEORETICAL MODEL

In the shear flow model, the shear velocity emerges as a natural governing parameter, which is a measure of shear stress at the wall. Shear velocity and flow angle measurements are made in order to quantify the Green's function in a local sense. Batchelder and Moffat [11] observed that the thermal wake from small sources of heat is an effective flow visualization tool, which suggests that thermal wake data from a small heat source may be used to determine the local shear velocity.

The process of using thermal response measurements to quantify convective heat transfer is diagrammed in Figure 1. Local measurements of temperature T_M from a small source of heat q_M are made at the boundary. These data are used to infer local values of shear velocity u_τ and flow angle β , which can be used to determine the local UTGF. A global shear velocity is constructed from local measurements in order to determine the global UTGF. Finally, the global UTGF is integrated over any heat source distribution to produce steady heat transfer data for use in conjugate analysis problems.

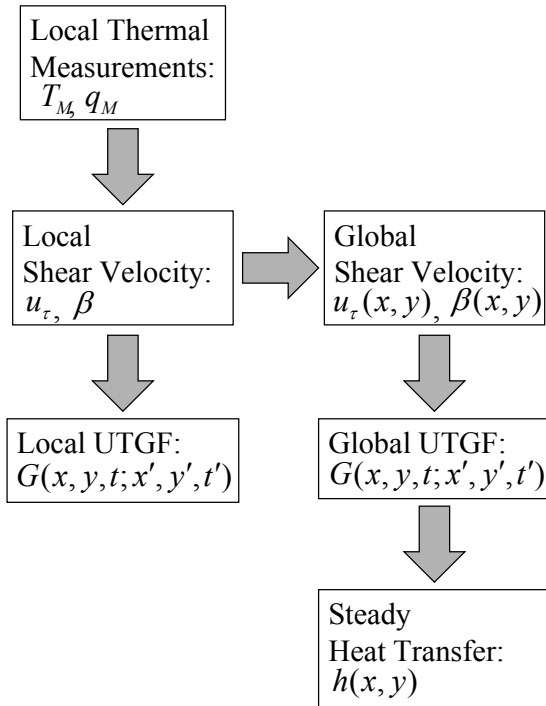


Fig 1: UTGF measurement process diagram

Limitations for the use of the linear shear flow model would ordinarily require the thermal boundary layer to not exceed the fluid boundary layer, which results in an upper bound on the Prandtl number for the laminar case, and a more severe restriction for the turbulent case. Extensions to more general cases may require a different thermo-fluid model, or perhaps by interpreting the shear velocity as a non-physical or “effective” quantity.

Thermo-Fluid Model

The geometry for the heat transfer problem of interest is shown in Figure 2. An arbitrary heat source $q(x, y, t)$ located on the boundary in a semi-infinite domain produces a temperature field $T(x, y, t)$ in the presence of a boundary layer flow. The transient convection-diffusion equation for this geometry near an adiabatic wall neglecting radiant emission and viscous dissipation can be written as

$$\frac{\partial T}{\partial t} + \mathbf{u}(x, y, z) \cdot \nabla T = \nabla \cdot (\varepsilon \nabla T) + \frac{1}{\rho c} q(x, y, t) \delta(z). \quad (1)$$

$$-k \frac{\partial T}{\partial z} \Big|_{z=0} = 0; \quad T(x, y, \infty, t) \rightarrow T_\infty$$

The diffusivity is interpreted as the sum of molecular and eddy diffusivities $\varepsilon = \alpha + \varepsilon_H$ for the turbulent boundary layer, or simply as the molecular diffusivity $\varepsilon = \alpha$ for the laminar boundary layer. The boundary conditions are an insulated wall condition and a freestream temperature far from the heat source.

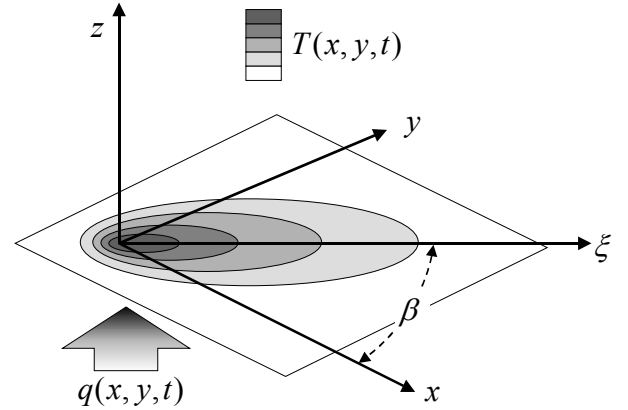


Fig 2: Configuration of thermo-fluid model

Two key assumptions are made. For small changes in temperature, the thermal properties of the fluid will be approximately constant, so that (1) becomes a linear equation, enabling formulation of the solution in terms of a Green's function

$$T(x, y, t) = T_\infty + \int_0^t dt' \int_{-\infty}^{\infty} dx' \int_{-\infty}^{\infty} dy' G(x, y, t; x', y', t') q(x', y', t'). \quad (2)$$

The $G(x, y, t; x', y', t')$ is the UTGF of interest. Here the convention adopted is that all quantities are evaluated at the boundary ($z = 0$) whenever the normal coordinate z is omitted.

Secondly, a simplified fluid model is utilized to simplify the analysis of the convective terms. The assumption is made that the majority of the convective effects take place within a narrow layer of fluid near the wall in which the velocity field may be approximated as unidirectional and linear in the normal coordinate. This formulation emerges as a near wall approximation in both laminar and turbulent boundary layer flows. In the case of the turbulent boundary layer, the near wall velocity field approximation for the viscous sublayer given by the “Law of the Wall” [12]

$$u(z) = \frac{u_\tau^2}{\nu} z, \quad (3)$$

where $u_\tau = \sqrt{\tau_0/\rho}$ is the shear velocity and τ_0 is the wall shear stress. For the turbulent boundary layer, the diffusivity reduces to the ordinary molecular diffusivity, i.e. $\varepsilon \equiv \alpha$ as in the laminar regime.

For the general 3-D formulation, a shear flow angle β is defined as the angle between the flow direction and some convenient coordinate in the plane of the wall (e.g. the x-coordinate in Figure 2). Local values $\beta(x, y)$ can be obtained from steady temperature response measurements $T(x', y')$ from a small circularly symmetric heat source located at (x, y) by finding a polar centroid, where (x', y') coordinates are centered about the heat source.

$$\beta(x, y) = \frac{\int_{-\infty}^{\infty} dx' \int_{-\infty}^{\infty} dy' \tan^{-1}(y'/x') \sqrt{x'^2 + y'^2} T(x', y')}{\int_{-\infty}^{\infty} dx' \int_{-\infty}^{\infty} dy' \sqrt{x'^2 + y'^2} T(x', y')} \quad (4)$$

If, in addition, local values $u_\tau(x, y)$ of the shear velocity are known, the general velocity field can be represented as
If, in addition, local values $u_\tau(x, y)$ of the shear velocity are known, the general velocity field can be represented as

$$\mathbf{u}(x, y, z) = \frac{u_\tau^2(x, y)}{\nu} z [\mathbf{e}_x \cos(\beta(x, y)) + \mathbf{e}_y \sin(\beta(x, y))] \quad (5)$$

Using this velocity field, the solution of (1) and (2) will yield a unique UTGF.

Local Shear Flow Approximation

For heat flux acting over a sufficiently small spatial domain, the flow parameters u_τ and β can be approximated as locally invariant, and enable (1) to be recast in a nondimensional form. Equation (5) suggests a change of coordinate system in order to simplify the convective terms. The energy equation is nondimensionalized using the following variables:

$$\xi = \frac{x \cos(\beta) + y \sin(\beta)}{\alpha Pr^{1/2}/u_\tau}, \quad \eta = \frac{-x \sin(\beta) + y \cos(\beta)}{\alpha Pr^{1/2}/u_\tau}, \quad (6)$$

$$\zeta = \frac{z}{\alpha Pr^{1/2}/u_\tau}, \quad \tau = \frac{t}{\alpha Pr/u_\tau^2}, \quad \theta = \frac{T - T_\infty}{Q Pr^{1/2}/\rho c u_\tau}, \quad \sigma = \frac{q}{Q}.$$

The variable Q represents some convenient heat flux scale, typically a maximum local value. The nondimensional form of the energy equation (1) is

$$\frac{\partial \theta}{\partial \tau} + \zeta \frac{\partial \theta}{\partial \xi} = \nabla^2 \theta + \sigma(\xi, \eta, \tau) \delta(\zeta). \quad (7)$$

A solution is found in terms of a nondimensional UTGF as follows,

$$\theta(\xi, \eta, \tau) = \int_0^\tau d\tau' \int_{-\infty}^\infty d\xi' \int_{-\infty}^\infty d\eta' \hat{G}(\xi, \eta, \tau; \xi', \eta', \tau') \sigma(\xi', \eta', \tau') \quad (8)$$

$$\frac{\partial \hat{G}}{\partial \tau} + \zeta \frac{\partial \hat{G}}{\partial \xi} = \nabla^2 \hat{G} + \delta(\xi - \xi', \eta - \eta', \zeta, \tau - \tau') \quad (9)$$

$$\left. \frac{\partial \hat{G}}{\partial \zeta} \right|_{\zeta=0} = 0$$

where $\delta(\xi, \eta, \zeta, \tau) \equiv \delta(\xi) \delta(\eta) \delta(\zeta) \delta(\tau)$.

In order to make the solution of (9) more tractable, two assumptions are made to simplify the functional form of $\hat{G}$. First, it is assumed that $\hat{G}$ does not depend on the temperature and source coordinates (spatial and temporal) independently, but rather on a difference. This is appropriate for a weakly varying flow field, which allows the use of Maxwell's reciprocity to rewrite $\hat{G}$ and θ as

$$\hat{G}(\xi, \eta, \tau; \xi', \eta', \tau') = \hat{G}(\xi - \xi', \eta - \eta', \tau - \tau'; 0, 0, 0) \quad (10)$$

$$\equiv \hat{G}(\xi - \xi', \eta - \eta', \tau - \tau')$$

$$\theta(\xi, \eta, \tau) = \int_0^\tau d\tau' \int_{-\infty}^\infty d\xi' \int_{-\infty}^\infty d\eta' \quad (11)$$

$$\times \hat{G}(\xi - \xi', \eta - \eta', \tau - \tau') \sigma(\xi', \eta', \tau').$$

Secondly, we assume a Gaussian distribution in the spanwise direction, as used by Smith [13]. This can be justified by inserting a trial solution (12) into (9) for $\tau > \tau'$

$$\hat{G}(\xi - \xi', \eta - \eta', \tau - \tau') = \quad (12)$$

$$= \frac{e^{-\frac{(\eta - \eta')^2}{4(\tau - \tau')}}}{\sqrt{4\pi(\tau - \tau')}} \hat{g}(\xi - \xi', \zeta - \zeta', \tau - \tau'),$$

which yields a reduced form for the UTGF solution equation.

$$\frac{\partial \hat{g}}{\partial t} + \zeta \frac{\partial \hat{g}}{\partial \xi} = \frac{\partial^2 \hat{g}}{\partial \xi^2} + \frac{\partial^2 \hat{g}}{\partial \zeta^2} + \delta(\xi - \xi', \zeta, \tau - \tau') ; \quad (13)$$

$$\left. \frac{\partial \hat{g}}{\partial \zeta} \right|_{\zeta=0} = 0$$

Equation (13) is solved numerically and interpolated between tabulated data points to construct UTGF solutions (12).

Finite Element Solution Kernel

A variety of numerical methods may be used to solve (13). One aspect of the problem makes the finite element method (FEM) particularly attractive: it provides a convenient way of modeling Dirac's delta function since it appears under an integral operator. By the selectivity property of the delta function, the resulting integral will have an analytical result. An existing 2-D transient Navier-Stokes solver was modified to include the impulsive source term as well as major simplifications to the velocity solver. This code, available from [14], uses a Petrov-Galerkin FEM, which is in general 3rd order accurate in space and 2nd order in time. Several irregular mesh domains with corresponding time step sizes were used in order to achieve better than order 1% accuracy.

For early time, (13) reduces to a pure Gaussian wave with no convection, which has a well-known analytical solution, within the accuracy requirements for $\tau \leq 3 \times 10^{-3}$.

$$\hat{g}(\xi, \zeta, \tau) \cong \frac{1}{2\pi\tau} \exp\left(-\frac{\xi^2 + \zeta^2}{4\tau}\right); \quad \tau \leq 3 \times 10^{-3} \quad (14)$$

Equation (14) is used in place of numerical data for early time, while tabulated numerical results are used elsewhere.

Accuracy of the method is verified by comparison with (14) for early time, and by using the FEM to solve an available analytic solution. Plots of (12) using numerical FEM solutions of (13) are shown in for various times in Figures 3-7.

UTGF MEASUREMENT THEORY

Equations (12) and (13) are converted back to dimensional form in order to analyze the temperature response T_M to a small circularly symmetric heat source q_M , both acquired from experimental measurements.

$$T(x, y, t) = T_\infty + \int_0^t dt' \int_{-\infty}^{\infty} dx' \int_{-\infty}^{\infty} dy' \times G(x - x', y - y', t - t') q_M(x', y', t') \quad (15)$$

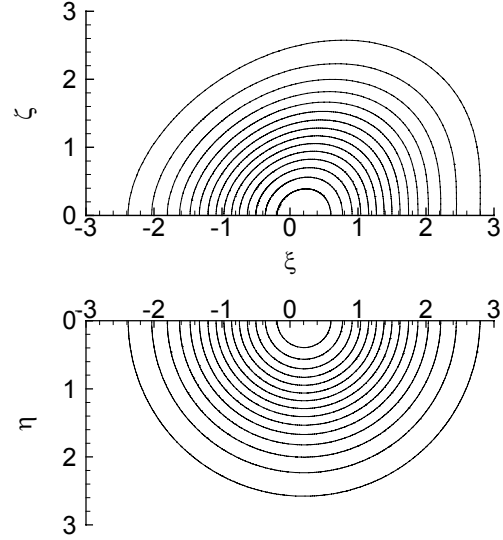


Fig. 3: Nondimensional temperature contours for the UTGF solution kernel; $\tau = 0.6$

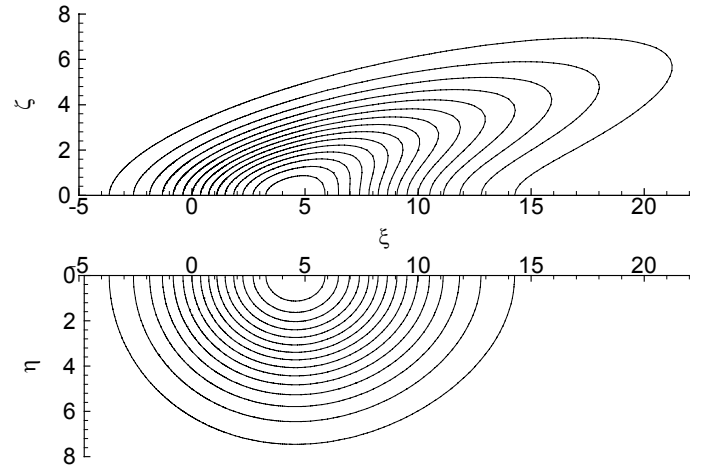


Fig. 4: Nondimensional temperature contours for the UTGF solution kernel; $\tau = 5$

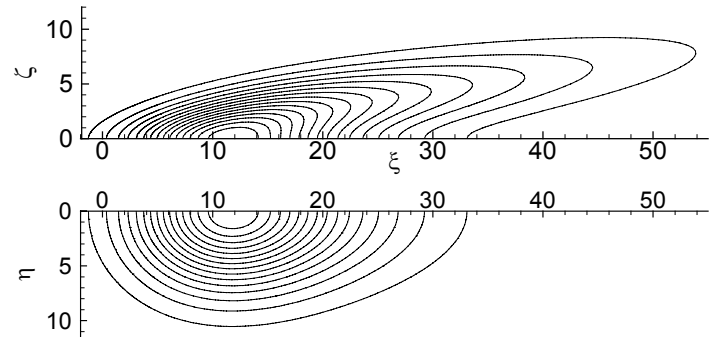


Fig. 5: Nondimensional temperature contours for the UTGF solution kernel; $\tau = 10$

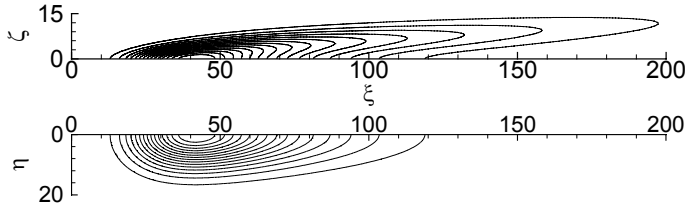


Fig. 6: Nondimensional temperature contours for the UTGF solution kernel; $\tau = 25$

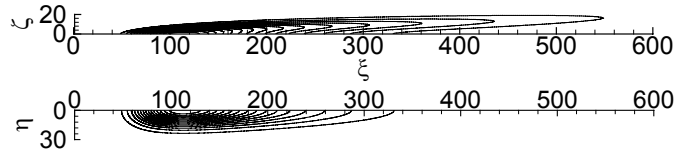


Fig. 7: Nondimensional temperature contours for the UTGF solution kernel; $\tau = 50$

$$G(x-x', y-y', t-t') = \frac{u_\tau^2}{\rho c \alpha^{5/2} Pr} \frac{\exp\left(-\frac{(y-y')^2}{4\alpha(t-t')}\right)}{\sqrt{4\pi(t-t')}} \times \hat{g}\left(\frac{(x-x')}{\alpha Pr^{1/2}/u_\tau}, \frac{t-t'}{\alpha Pr/u_\tau^2}\right) \quad (16)$$

Specification of a heat source q_M introduces a length scale (D_M), typically taken to be a diameter. This length scale, nondimensionalized with respect to the same scale length scale given by (6), yields the governing parameter for the solution as $Pe_{\tau,M} Pr^{-1/2}$ in terms of a shear Peclet number defined by

$$Pe_{\tau,M} = \frac{u_\tau D_M}{\alpha} \quad (17)$$

Computed (15) and measured (T_M) temperature responses are compared using an error measure defined as

$$E(u_\tau) = \left[\int_0^\infty dt' \int_{-\infty}^\infty dx' \int_{-\infty}^\infty dy' (T - T_M)^2 \right]^{1/2} \quad (18)$$

This function is minimized at each boundary point of interest to determine an effective u_τ characterizing the flow.

In (18), the spatial domain is given as the entire boundary, but in practice, only a portion may need to be measured. In principle, the temporal evolution of the maximum temperature and position from a pulse heat source in time is sufficient to extract the UTGF. However, more evaluation points used to evaluate (18) may yield a more robust method.

The calculated shear velocities, in addition to shear flow angle measurements made using (3), are interpolated and used to determine a global temperature response and UTGF.

$$T(x, y, t) = T_\infty + \int_0^t dt' \int_{-\infty}^\infty dx' \int_{-\infty}^\infty dy' \times G(x, y, t; x', y', t') q(x', y', t') \quad (19)$$

$$G(x, y, t; x', y', t') = \frac{u_\tau^2(x', y')}{\rho c \alpha^{5/2} Pr} \exp\left(-\frac{[-(x-x')\sin(\beta(x', y')) + (y-y')\sin(\beta(x', y'))]^2}{4\alpha(t-t')}\right) \times \frac{1}{\sqrt{4\pi(t-t')}} \times \hat{g}\left(\frac{(x-x')\cos(\beta(x', y')) + (y-y')\sin(\beta(x', y'))}{\alpha Pr^{1/2}/u_\tau(x', y')}, \frac{t-t'}{\alpha Pr/u_\tau^2(x', y')}\right) \quad (20)$$

Note that Maxwell's reciprocity is no longer valid, since $u_\tau(x, y)$ and $\beta(x, y)$ are no longer constant.

Where a convenient length scale D can be specified for the heat source in (19), a global Peclet number can be specified as

$$Pe_\tau = \frac{u_\tau D}{\alpha} \quad (22)$$

Finally, if desired, heat transfer coefficient data for a steady source $q(x, y)$ is obtained by integrating the UTGF

$$h(x, y) = \frac{q(x, y)}{T(x, y) - T_\infty} = \frac{q(x, y)}{\int_0^\infty dt' \int_{-\infty}^\infty dx' \int_{-\infty}^\infty dy' G(x-x', y-y', t-t') q(x', y')} \quad (22)$$

PRELIMINARY EXPERIMENTAL INVESTIGATION

The experimental investigation is in the preliminary stage, but will entail simultaneous measurements of the transient heat flux distribution and temperature response data on an adiabatic boundary. Heat flux application will be made using an impinging laser source of known distribution. Details of the experiment will be adjusted as needed in response to future needs, although two issues are anticipated: specification of the size and functional form of the heat flux distribution, and proper selection of adiabatic boundary material.

The functional form of the heat flux distribution used to measure the shear velocity field may be subject to the availability of laser sources. Axisymmetry is necessary in order

to obtain β measurements using (3), but as this is only required for β , a separate non-axisymmetric heat source may be used for u_τ measurements as long as the heat flux distribution is known. The temporal evolution of the source can be chosen to maximize the robustness of the UTGF measurement technique.

Sizing issues for the heat source is based on the contrasting needs of displaying a noticeable thermal wake, while being sufficiently small to be interpreted as pointwise data. The first consideration concerns the parameter $Pe_{\tau,M} Pr^{-1/2}$. The thermal wake effect will be more noticeable for certain values of this parameter, and this places a lower bound on the source size required for measurement. The upper bound is related to the assumption of a locally constant shear velocity. For highest accuracy the source size should be as small as possible subject to the first consideration.

Boundary material will need to be carefully chosen in order to keep the adiabatic wall assumption and avoid the full conjugate problem. In the preliminary temperature data, conjugate effects are noticeable, which suggests this will be a difficult requirement to meet.

Preliminary experimental data were collected from steady UTGF temperature measurements in a fully developed channel flow using a heat source conducting through a thin wire [15]. Thermal response data was taken from thermochromic liquid crystal visualized using a digital imaging station. At present, this is a proof-of-concept experiment mainly for the purposes of anticipating concerns for future generations of the experiment. Typical results are shown for both a laminar and turbulent case in Figures 8 and 9. These images have been digitally retouched, and the white circles denote the approximate size of the heat source.

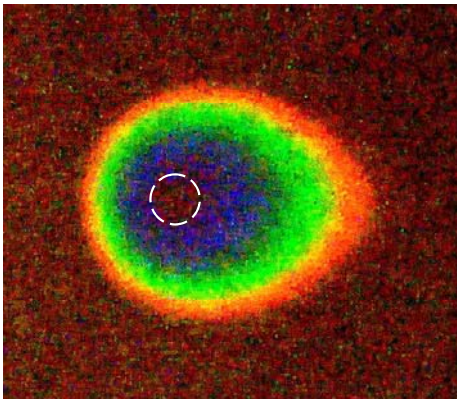


Fig. 8: Laminar channel flow TLC image data;
 $Pe_{\tau,M} Pr^{-1/2} = 11.8$

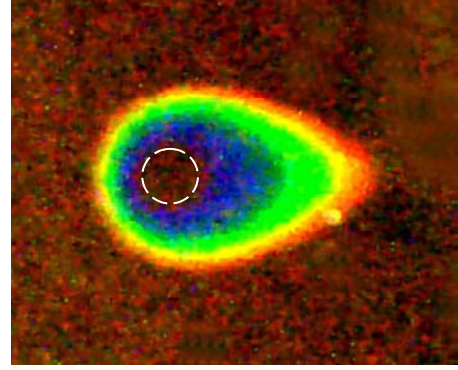


Fig. 9: Turbulent channel flow TLC image data;
 $Pe_{\tau,M} Pr^{-1/2} = 83$

The UTGF solution for the laminar flow is found by integrating (10) over a circular heat source to steady state. This is shown in Figure 10, with the dotted black line denoting the size of the heat source. Comparing Figures 8 and 10, there is noticeable difference between the thermal plumes, which shows evidence of substantial conjugate effects.

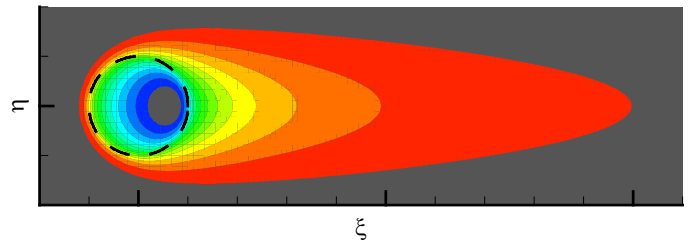


Fig. 10: Integrated UTGF over circular source of heat;
 $Pe_{\tau,M} Pr^{-1/2} = 11.8$

CONCLUSIONS

The unsteady thermal Green's function (UTGF) measurement theory was presented as a new approach to use transient thermal response data to infer a Green's function in a 3-D boundary layer. Using a linear shear flow model, the Green's function solution is parameterized in terms of local shear velocity and flow angle measurements. Such measurements, in addition to FEM solutions to the nondimensional energy equation are used to construct a Green's function to specify the relationship between wall heat flux and temperature response. This Green's function can be integrated to produce steady heat transfer data.

Experimental verification requires simultaneous transient temperature and heat flux measurements. Future steps in the experimental implementation of this theory will address issues such as: dependence of the size and temporal response of the laser heat source on the measurement accuracy, selection of an

adiabatic wall material, and development of improved temperature measurement techniques.

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REFERENCES

- [1] Moffat, R. J., 1998, "What's new in convective heat transfer?" *Int. J. Heat and Fluid Flow*, Vol. 19, pp. 90-101.
- [2] Ortega, A., and Ramanathan, S., 2003, "On the Use of Point Source Solutions for Forced Air Cooling of Electronic Components—Part I: Thermal Wake Models for Rectangular Heat Sources," *ASME J. Electronics Packaging*, Vol. 125, No. 2, pp. 226-234.
- [3] Ortega, A., and Ramanathan, S., 2003, "On the Use of Point Source Solutions for Forced Air Cooling of Electronic Components—Part II: Conjugate Forced Convection From a Discrete Rectangular Source on a Thin Conducting Plate," *ASME J. Electronics Packaging*, Vol. 125, No. 2, pp. 235-243.
- [4] Li, Y., 1997, "Thermal Analysis for Convective Cooling of Electronic Components Using a Shear Flow Model," Masters thesis, *University of Arizona, Dept. of Aerospace and Mechanical Engineering*.
- [5] Cole, K. D., and Beck, J. V., 1988, "Conjugated Heat Transfer from a Strip Heater with the Unsteady Surface Element Method," *J. Thermophysics*, Vol. 1, No. 4, pp.348-354.
- [6] Liang, P. W., and Cole, K. D., 1992, "Transient Conjugated Heat Transfer From a Rectangular Hot Film," *J. Thermophysics*, Vol. 6, No. 2, pp.349-355.
- [7] Batchelder, K. A., and Eaton, J. K., 2001, "Practical Experience With the Discrete Green's Function Approach to Convective Heat Transfer," *ASME J. Heat Transfer*, Vol. 123, pp. 70-76.
- [8] Hacker, J. M., and Eaton, J. K., 1995, "Heat Transfer Measurements in a Backward Facing Step Flow with Arbitrary Wall Temperature Variations," Thermosciences Division Report MD-71, Mechanical Engineering Department, Stanford University, Stanford, CA.
- [9] Mukerji, D., and Eaton, J. K., 2002, "Spatially-Resolved Measurements of Heat Transfer in Turbomachinery Applications," Thermosciences Division Report TSD-145, Mechanical Engineering Department, Stanford University, Stanford, CA.
- [10] Batchelder, K. A., and Moffat, R. J., 1997, "Towards a Method for Measuring Heat Transfer in Complex 3-D Flows," Thermosciences Division Report TSD-108, Mechanical Engineering Department, Stanford University, Stanford, CA.
- [11] Batchelder, K. A., and Moffat, R. J., 1998, "Surface Flow Visualization Using the Thermal Wakes of Small Heated Spots," *Experiments in Fluids*, Vol. 25, pp.104-107.
- [12] Kays, W. M., and Crawford, M. E., 1993, *Convective Heat and Mass Transfer*, 3rd Edition, McGraw-Hill, New York, NY, pp. 198-213.
- [13] Smith, F. B., 1957, "The Diffusion of Smoke From a Continuous Elevated Point-Source Into a Turbulent Atmosphere," *J. Fluid Mechanics*, Vol. 2, pp 49-76.
- [14] Heinrich, H. C., and Pepper, D. W., 1999, *Intermediate Finite Element Method: Fluid Flow and Heat Transfer Applications*, Taylor & Francis, Philadelphia, PA, pp. 357-402.
- [15] Ortega, A., Wodrich, J., and Radtke, G., 2003, "Experimental Measurement of Steady 3-D Thermal Green's Functions," 2003 *ASME International Mechanical Engineering Congress and Exposition*, Washington D. C., Nov. 5-11 2003.